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Inventor(s)

Song; Hae Lyong et al.

MEMORY SYSTEM ADAPTIVELY ALLOCATING A BUFFER MEMORY, MEMORY CONTROLLER AND METHOD OF OPERATING THE SAME

Abstract

A memory system may include a storage device, a buffer memory and a memory controller. The buffer memory may include a first portion and a second portion. The memory controller may detect a set event, store external data in the first portion before the event is detected, and store the external data in the second portion after the event is detected. The external data is associated with a read request or a write request from an external device.

Inventors: Song; Hae Lyong (Gyeonggi-do, KR), SHIN; Woong Sik (Gyeonggi-do, KR)

Applicant: SK hynix Inc. (Gyeonggi-do, KR)

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 18/466,831 filed on Sep. 14, 2023, which claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2023-0054108, filed on Apr. 25, 2023, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] Various embodiments of the present disclosure generally relate to a semiconductor integrated device, particularly, to a memory system adaptively allocating a buffer memory, a memory controller and a method of operating the same.

2. Related Art

[0003] A computer system may use a buffer memory for reducing a bottle neck caused by a difference in an operation speed between a data transmission device and a data reception device.

[0004] A memory system may perform data input/output operations using a volatile or non-volatile memory device in accordance with requests from an external device. For example, in order to prevent loss of data, the memory system may use the buffer memory to compensate for the difference in data transmission speeds between the data transmission device and the data reception device.

[0005] Therefore, it may be required to provide the memory system with the buffer memory having a bandwidth and a capacity which can secure optimal data transmission performance.

SUMMARY

[0006] According to example embodiments of the present disclosure, there may be provided a memory system. The memory system may include a storage device; a buffer memory including a first portion and a second portion; and a memory controller suitable for: detecting a set event, storing external data in the first portion before the event is detected, and storing the external data in the second portion after the event is detected, wherein the external data is associated with a read request or a write request from an external device.

[0007] According to example embodiments of the present disclosure, there may be provided a memory controller. The memory controller may include a first memory device; a second memory device; and a processor suitable for: detecting an event, which is triggered by at least one of an available space capacity of a storage device and an aging detection signal of the storage device, before the event is detected, storing, in the first memory device, external data read-requested or write-requested by an external device, and after the event is detected, storing, in the memory devices, the external data, which is transferred regardless of the read request or the write request from the external device.

[0008] According to example embodiments of the present disclosure, there may be provided a method of operating a memory system. The method of operating a memory system which includes a storage device and a memory controller controlling a buffer memory including a first portion and a second portion, the method comprising: detecting, by the memory controller, a set event; before the event is detected, storing, by the memory controller, in the first portion, external data, which is associated with a read request or a write request from an external device; and after the event is detected, storing, by the memory controller, the external data in the second portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The above and another aspects, features and advantages of the subject matter of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0010] FIG. 1 is a diagram illustrating a memory system in accordance with example embodiments of the present disclosure;

[0011] FIG. 2 is a diagram illustrating a data transmission concept in a memory system in accordance with example embodiments of the present disclosure;

[0012] FIG. 3 is a diagram illustrating a buffer memory arbitration concept based on an event detection in accordance with example embodiments of the present disclosure;

[0013] FIG. 4 is a diagram illustrating a buffer memory arbitration concept based on an event detection in accordance with example embodiments of the present disclosure; and

[0014] FIG. 5 is a flow chart illustrating a method of operating a memory system in accordance with example embodiments of the present disclosure.

DETAILED DESCRIPTION

[0015] Various embodiments of the present invention will be described in greater detail with reference to the accompanying drawings. The drawings are schematic illustrations of various embodiments (and intermediate structures). As such, variations from the configurations and shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, the described embodiments should not be construed as being limited to the particular configurations and shapes illustrated herein but may include deviations in configurations and shapes which do not depart from the spirit and scope of the present invention as defined in the appended claims.

[0016] The present invention is described herein with reference to cross-section and/or plan illustrations of embodiments of the present invention. However, embodiments of the present invention should not be construed as limiting the inventive concept. Although a few embodiments of the present invention will be shown and described, it will be appreciated by those of ordinary skill in the art that changes may be made in these embodiments without departing from the principles and spirit of the present invention.

[0017] FIG. 1 is a diagram illustrating a memory system **10** in accordance with example embodiments of the present disclosure.

[0018] Referring to FIG. 1, the memory system **10** may include a memory controller **100** and a storage device **200**. The memory system may be electrically connected with an external device to exchange data with the external device.

[0019] In some embodiments, the external device may include a personal computer, a laptop computer, a server computer, a work station, a tablet PC, a drone, an advanced drivers assistance system (ADAS), a smart TV, a smart phone, a medical device, an image display, a calibrator, an IoT, etc.

[0020] The storage device **200** may include at least one of a volatile memory device and a non-volatile memory device. The storage device **200** may be electrically connected with the memory controller **100** via a plurality of channels CH1~CHm. In the illustrated embodiments, the storage device **200** may include a plurality of memory chips CHIP. Alternatively, the storage device **200** may include a plurality of packages. Each memory chip CHIP may include a plurality of dies. Each of the dies may include at least one of a plurality of planes. Each of the planes may include at least one of a plurality of memory blocks. Each of the memory blocks may include a plurality of pages. A physical address may be allotted to a memory region of the storage device **200**, for example, a page, a memory block and a die.

[0021] The memory controller **100** may control the storage device **200** in response to a request from the external device. For example, the memory controller **100** may program data in the storage device **200** in accordance with a write request from the external device. The memory controller **100** may read the data in the storage device **200** in response to a read request from the external device. The memory controller **100** may then provide the external device with the read data.

[0022] The memory controller **100** may include at least one of a plurality of processors **111**, an external interface **113**, a storage interface **115**, a buffer manager **117**, a first memory device **121** and a second memory device **123**. The first memory device **121** and the second memory device **123** may be implemented with a buffer memory.

[0023] The processor **111** may include a combination of a hardware and firmware (FW) or a software operated on the hardware. The hardware may be operated by performing the firmware or the software for operating the memory system **10**. In example embodiments, the processor **111** may include a flash transition layer (FTL) for managing the storage device **200**, which is used to perform, for example, a garbage collection, an address mapping, a wear leveling, etc.

[0024] The external interface **113** may provide communication channels for receiving a command and a clock signal from the external device and controlling input/output of data in accordance with a control of the processor **111**. The external interface **113** may provide a physical connection between the external device and the memory system **10**.

[0025] The external interface **113** may temporarily store write data, which may be provided from the external device in accordance with the control of the processor **111**, in one of the first memory device **121** and the second memory device **123**. The external interface **113** may provide the external device with read data temporarily stored in the buffer memory **121** or **123**.

[0026] In example embodiments, the external interface **113** may communicate with the external device based on a communication standard or interface using at least one of a USB protocol, a multimedia card protocol, PCI (peripheral component interconnection) protocol, PCI-e or PCIe (PCI-express) protocol, ATA (Advanced Technology Attachment) protocol, SATA (Serial-ATA) protocol, PATA (Parallel-ATA) protocol, SCSI (small computer small interface) protocol, ESDI (enhanced small disk interface) protocol, IDE (Integrated Drive Electronics) protocol, private protocol, SMBus (System Management Bus) protocol, I2C (Inter-Integrated Circuit) protocol, I3C (Improved Inter-Integrated Circuit) protocol, etc.

[0027] The storage interface **115** may provide communication channels between the memory controller **100** and the storage device **200**. The storage interface **115** may register data, which may be temporarily stored in the buffer memory **121** or **123**, in the storage device **200** in accordance with the control of the processor **111**. The storage interface **115** may transmit the data read from the storage device **200** to the buffer memory **121** and **123**. The read data may be temporarily stored in the buffer memory **121** or **123**.

[0028] The buffer manager **117** may control the second memory device **123** in accordance with the control of the processor **111**.

[0029] In the illustrated embodiment, the first memory device **121** may be positioned inside the memory controller **100**, the second memory device **123** may be positioned outside the memory controller **100**.

[0030] In example embodiments, the first memory device **121** may have an input/output bandwidth greater than an input/output bandwidth of the second memory device **123**. The first memory device **121** may have an operation speed faster than that of the second memory device **123**. The first memory device **121** may include a static random access memory (SRAM). The second memory device **123** may include a dynamic random access memory (DRAM). However, the characteristics of the first memory device **121** and the second memory device **123** may not be restricted within the above-mentioned characteristics.

[0031] The write data may be provided from the external device in accordance with the write request from the external device. The write data may be programmed in the storage device **200** by

the control of the memory controller **100**. The read data may be read from the storage device **200** by the control of the memory controller **100** in accordance with the read request of the external device. The read data may be provided to the external device. The write data and the read data may correspond to the external data.

[0032] Since a read operation, a write operation and an erase operation may be repeated, a storage space of the storage device **200** may be reduced and aged. In order to secure an available space of the storage device **200** or uniformly use blocks in the storage device **200**, the memory controller **100** may perform storage management such as a garbage collection, a wear leveling, a read reclaim, etc. The storage management may be performed by an internal operation such as a background operation regardless of the request from the external device. Data transmitted and/or received between the memory controller **100** and the storage device **200** regardless of the read request or the write request from the external device by the background operation may correspond to the internal data.

[0033] The processor **111** may determine a work load pattern in accordance triggering of a set event. The processor **111** may arbitrate a buffering of the external data and the internal data in one of the buffer memories **121** and **123**.

[0034] In example embodiments, the set event may be triggered based on at least one of an available space capacity of the storage device **200** and an aging detection signal. The available space may be an empty memory region, for example, a free block. The aging detection signal may represent an aging state of the memory system **10**. When the storage management is required, the aging detection signal may be activated by the processor **111**. Specific conditions and a method of triggering the set event will be described later.

[0035] FIG. **2** is a diagram illustrating a data transmission concept in a memory system in accordance with example embodiments of the present disclosure.

[0036] Referring to FIG. **2**, the processor **111** may include a block manager **201** and an arbiter **203**.

[0037] The external interface **113** may include an external interface buffer (E_BUFFER) **1131**. The storage interface **115** may include a storage interface buffer (S_BUFFER) **1151**.

[0038] The write data provided from the external device in accordance with the write request from the external device may be temporarily stored in the external interface buffer **1131**. The write data may be transmitted to a specific portion of the buffer memory **121** and **123** allotted by the processor **111**. The write data in the specific portion of the buffer memory **121** and **123** may be provided to the storage device **200** through the storage interface buffer **1151**.

[0039] The read data provided from the external device in accordance with the read request from the external device may be temporarily stored in the storage interface buffer **1151**. The read data may be transmitted to a specific portion of the buffer memory **121** and **123** allotted by the processor **111**. The read data in the specific portion of the buffer memory **121** and **123** may be provided to the external device through the external interface buffer **1131**.

[0040] The block manager **201** may manage state information of the memory blocks in the storage device **200**. For example, the block manager **201** may manage the available space capacity, a wear volume, free block information, bad block information, etc.

[0041] The block manager **201** may trigger the set event based on the state information of the memory block.

[0042] In some embodiments, the block manager **201** may trigger the set event based on at least one of the available space capacity of the storage device **200** and the aging detection signal.

[0043] For example, the block manager **201** may calculate the available space capacity corresponding to a number of the free blocks. The block manager **201** may determine a requirement of the storage management based on the wear volume and/or the number of the free blocks. When the storage management is required, the block manager **201** may activate the aging detection signal.

[0044] The arbiter **203** may determine one of the first memory device **121** or the second memory device **123** as a position where the external data and/or the internal data is to be buffered in

accordance with the triggering of the set event.

[0045] The block manger **201** may trigger the set event based on following conditions, not limited thereto. [0046] First Condition: the available space capacity of the storage device is no more than a first threshold value [0047] Second Condition: a ratio of the available space capacity with respect to a total space of the storage device is no more than a second threshold value [0048] Third Condition: at least one of the first and second conditions is satisfied [0049] Fourth Condition: the aging detection signal is once activated [0050] Fifth Condition: the aging detection signal is activated by no less than a reference number being greater than 1 [0051] Sixth Condition: the third condition and the fourth condition are satisfied [0052] Seventh Condition: the third condition and the fifth condition are satisfied

[0053] Before the set event is triggered, the arbiter **203** may control the external interface **113** to buffer the external data in the first memory device **121**.

[0054] After the set event is triggered, the arbiter **203** may control the external interface **113** to buffer the external data in the second memory device **123**. The arbiter **203** may control the storage interface **115** to buffer the internal data in the first memory device **121**.

[0055] Before the set event is triggered, the internal data may be generated or not. For example, in the first condition, the second condition, the third condition, the fifth condition and the seventh condition, the internal data may be generated before the set event is triggered. The arbiter **203** may control the storage interface **115** to buffer the internal data in the second memory device **123**.

[0056] In the fourth condition and the sixth condition, the internal data may not be generated before the set event is triggered. The buffering position of the internal data may be selectively selected from the first memory device **121** or the second memory device **123**, not limited thereto.

[0057] The buffering position of the internal data before triggering the set event may not be limited. The buffering position of the internal data before triggering the set event may be determined, as at least one of the first memory device **121** and the second memory device **123**, based on used states of the first memory device **121** or the second memory device **123**, a size of the internal data, etc.

[0058] FIG. **3** is a diagram illustrating a buffer memory arbitration concept based on an event detection in accordance with example embodiments of the present disclosure. In FIG. **3**, transmission paths of the external data EDATA and the internal data IDATA in accordance with the control of the processor **111** are shown before the set event is triggered.

[0059] The external data EDATA may include the write data and the read data transferred in accordance with the request from the external device. The internal data IDATA may be transferred between the memory controller **100** and the storage device **200** regardless of the request from the external device.

[0060] When the set event is not triggered by the block manager **201**, the arbiter **203** may transmit an arbitration signal ABT to the external interface **113** and the storage interface **115** to control the buffering position of the external data EDATA and the internal data IDATA.

[0061] The external interface **113** may store the external data EDATA, for example, the write data, in the first memory device **121** in response to the arbitration signal ABT.

[0062] The storage interface **115** may store the external data EDATA, for example, the read data, which has been stored in the storage interface buffer **1151**, in the first memory device **121** in response to the arbitration signal ABT. The storage interface **115** may store the internal data IDATA, which has been stored in the storage interface buffer **1151**, in the second memory device **123** in response to the arbitration signal ABT. The buffering position of the internal data IDATA may not be limited. The buffering position of the internal data IDATA may be determined, as at least one of the first memory device **121** and the second memory device **123**, based on used states of the first memory device **121** or the second memory device **123**, a size of the internal data, etc.

[0063] For example, the arbiter **203** may determine the buffering position of the internal data IDATA in accordance with at least one of the used states of the first memory device **121** or the second memory device **123** and the size of the internal data, and may then generate the arbitration

signal ABT.

[0064] That is, before the set event is triggered, the arbiter **203** may control the external interface **113** and the storage interface **115** to buffer the external data EDATA in the first memory device **121**.

[0065] Further, before the set event is triggered, the arbiter **203** may control the storage interface **115** to buffer the internal data IDATA in at least one of the first memory device **121** and the second memory device **123**.

[0066] FIG. **4** is a diagram illustrating a buffer memory arbitration concept based on an event detection in accordance with example embodiments of the present disclosure. In FIG. **4**, transmission paths of the external data EDATA and the internal data IDATA in accordance with the control of the processor **111** are shown after the set event is triggered.

[0067] The external data EDATA may include the write data and the read data. The internal data IDATA may include data transmitted by the background operation.

[0068] After the set event is triggered by the block manager **201**, the arbiter **203** may transmit the arbitration signal ABT to the external interface **113** and the storage interface **115** to control the buffering position of the external data EDATA and the internal data IDATA.

[0069] The external interface **113** may store the external data EDATA, for example, the write data, which has been stored in the external interface buffer **1131**, in the second memory device **123** in response to the arbitration signal ABT. The storage interface **115** may store the internal data IDATA, which has been stored in the storage interface buffer **1151**, in the first memory device **121** in response to the arbitration signal ABT.

[0070] That is, after the set event is triggered by the block manager **201**, the arbiter **203** may control the external interface **113** and the storage interface **115** to buffer the external data EDATA in the second memory device **123** and the internal data IDATA in the first memory device **121**.

[0071] FIG. **5** is a flow chart illustrating a method of operating a memory system in accordance with example embodiments of the present disclosure.

[0072] Referring to FIG. **5**, in operation **S101**, the memory controller **100** of the memory system **10** may be in a standby mode.

[0073] In operation **S103**, the memory controller **100** may monitor whether the set event is triggered or not. In one aspect, the memory controller **100** may monitor a triggered record or history indicating whether the set event has ever been triggered.

[0074] In example embodiments, when any of the above-mentioned seven conditions is satisfied, the set event may be triggered based on at least one of the available space capacity of the storage device **200** and the aging detection signal.

[0075] Before the set event is triggered, i.e., the set event does not have a triggered record, in operation **S105**, the memory controller **100** may control the external interface **113** and the storage interface **115** to buffer the external data in the first portion of the buffer memory, for example, the first memory device **121**. The memory controller **100** may continuously monitor the triggering of the set event.

[0076] Before the set event is triggered, i.e., the set event does not have a triggered record, in operation **S107**, the memory controller **100** may control the storage interface **115** to buffer the internal data in the first portion or the second portion of the buffer memory, for example, the first memory device **121** or the second memory device **123**. The memory controller **100** may continuously monitor the triggering of the set event.

[0077] After the set event is triggered, i.e., the set event has a triggered record, in operation **S109**, the memory controller **100** may control the external interface **113** and the storage interface **115** to buffer the external data in the second portion of the buffer memory, for example, the second memory device **123**.

[0078] After the set event is triggered, i.e., the set event does not have a triggered record, in operation **S111**, the memory controller **100** may control the storage interface **115** to buffer the

internal data in the first portion of the buffer memory, for example, the first memory device 121.

[0079] In the memory system such as an SSD for a high performance server with larger capacity, sequential write operations may be mainly used at an initial stage of the memory system. After the memory system has entered into a sustain mode by repeatedly using the memory system, the work load pattern mainly including a random write operation may be represented.

[0080] Therefore, in order to improve the processing capacity of the external data, the first memory device, which has the bandwidth and the operation speed greater than those of the second memory device, may be used as the buffer memory for the external data in the initial stage where the internal data by the background operation is not generated.

[0081] After the memory system has entered into the sustain mode, the first memory device may be used as the buffer memory for the internal data and the second memory device may be used as the buffer memory for the external data, in order to guarantee the required capacity of the external device without increasing the expensive first memory device.

[0082] As a result, the buffer memory may be arbitrated in accordance with the work load pattern to secure the data processing capacity and to reduce a cost for securing the buffer memory.

[0083] The above described embodiments of the present invention are intended to illustrate and not to limit the present invention. Various alternatives and equivalents are possible. The invention is not limited by the embodiments described herein. Nor is the invention limited to any specific type of semiconductor device. Other additions, subtractions, or modifications which are apparent in view of the present disclosure are intended to fall within the scope of the appended claims. Furthermore, the embodiments may be combined to form additional embodiments.

Claims

1. A memory controller comprising: a processor configured to detect an event; an arbiter configured to allocate a buffer memory including a first portion and a second portion; wherein the arbiter is configured to allocate the first portion to external data before the event is detected, without storing the external data in the second portion, and allocate the second portion to the external after the event is detected, without storing the external data in the first portion.
2. The memory controller of claim 1, the external data is associated with a read request or a write request from an external device.
3. The memory controller of claim 1, wherein the first portion has a bandwidth greater than that of the second portion.
4. The memory controller of claim 1, wherein the first portion is arranged inside the memory controller; and the second portion is arranged outside the memory controller.
5. The memory controller of claim 1, wherein the arbiter is configured to allocate the first portion to internal data.
6. The memory controller of claim 5, wherein the memory controller is configured to control a storage device, and the internal data is transferred between the memory controller and the storage device regardless of the read request or the write request from an external device.
7. The memory controller of claim 5, wherein the memory controller is configured to control a storage device, and the internal data comprises data associated with a background operation performed on the storage device.
8. The memory controller of claim 1, wherein the memory controller is configured to control a storage device, and the event is triggered based on an available space capacity of the storage device.
9. The memory controller of claim 1, wherein the memory controller is configured to control a storage device, and the event is triggered when a background operation performed on the storage device is activated.
10. A memory controller comprising: a processor configured to monitor an event is being triggered;

an arbiter configured to store external data to a second portion of a buffer memory without storing the external data in a first portion of the buffer memory when a condition to trigger the event is satisfied, and to store external data to the first portion without storing the external data in the second portion when the condition is not satisfied.

11. The memory controller of claim 10, the external data is associated with a read request or a write request from an external device.

12. The memory controller of claim 10, wherein the first portion has a bandwidth greater than that of the second portion.

13. The memory controller of claim 10, wherein the first portion is arranged inside the memory controller; and the second portion is arranged outside the memory controller.

14. The memory controller of claim 10, wherein the arbiter is configured to allocate the first portion to internal data.

15. The memory controller of claim 10, wherein the memory controller is configured to control a storage device, and the event is triggered based on an available space capacity of the storage device.

16. The memory controller of claim 10, wherein the memory controller is configured to control a storage device, and the event is triggered when a background operation performed on the storage device is activated.

17. A memory system comprising: a storage device; a first memory device; a second memory device; a processor configured to detect an event associated with the storage device; and an arbiter configured to allocate the first memory device to external data before the event is detected, without storing the external data in the second memory device, and allocate the second memory device to the external after the event is detected, without storing the external data in the first memory device.

18. The memory system of claim 17, wherein the first memory device has a bandwidth greater than that of the second memory device.

19. The memory system of claim 17, wherein the arbiter is configured to allocate the first portion to internal data.

20. The memory system of claim 17, wherein the event is triggered based on an available space capacity of the storage device or when a background operation performed on the storage device is activated.
